

11.4 Box Dimension

To deal with fractals that are not self-similar, we need to generalize our notion of dimension still further. Various definitions have been proposed; see Falconer (1990) for a lucid discussion. All the definitions share the idea of “measurement at a scale ε ”—roughly speaking, we measure the set in a way that ignores irregularities of size less than ε , and then study how the measurements vary as $\varepsilon \rightarrow 0$.

Definition of Box Dimension

One kind of measurement involves covering the set with boxes of size ε (Figure 11.4.1).

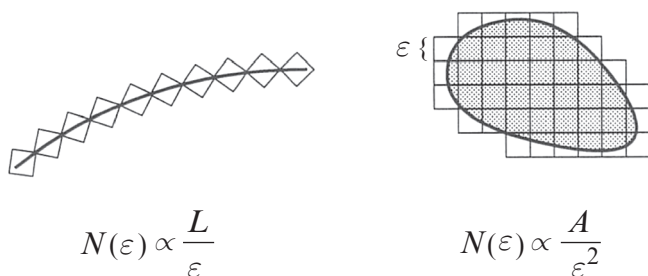


Figure 11.4.1

Let S be a subset of D -dimensional Euclidean space, and let $N(\varepsilon)$ be the minimum number of D -dimensional cubes of side ε needed to cover S . How does $N(\varepsilon)$ depend on ε ? To get some intuition, consider the classical sets shown in Figure 11.4.1. For a smooth curve of length L , $N(\varepsilon) \propto L/\varepsilon$; for a planar region of area A bounded by a smooth curve, $N(\varepsilon) \propto A/\varepsilon^2$. The key observation is that the dimension of the set equals the exponent d in the **power law** $N(\varepsilon) \propto 1/\varepsilon^d$.

This power law also holds for most fractal sets S , except that d is no longer an integer. By analogy with the classical case, we interpret d as a dimension, usually called the *capacity* or **box dimension** of S . An equivalent definition is

$$d = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)}, \text{ if the limit exists.}$$

EXAMPLE 11.4.1:

Find the box dimension of the Cantor set.

Solution: Recall that the Cantor set is covered by each of the sets S_n used in its construction (Figure 11.2.1). Each S_n consists of 2^n intervals of length $(1/3)^n$, so if we pick $\varepsilon = (1/3)^n$, we need all 2^n of these intervals to cover the Cantor set. Hence $N = 2^n$ when $\varepsilon = (1/3)^n$. Since $\varepsilon \rightarrow 0$ as $n \rightarrow \infty$, we find

$$d = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)} = \frac{\ln(2^n)}{\ln(3^n)} = \frac{n \ln 2}{n \ln 3} = \frac{\ln 2}{\ln 3}$$

in agreement with the similarity dimension found in Example 11.3.1. ■

This solution illustrates a helpful trick. We used a discrete sequence $\varepsilon = (1/3)^n$ that tends to zero as $n \rightarrow \infty$, even though the definition of box dimension says that we should let $\varepsilon \rightarrow 0$ continuously. If $\varepsilon \neq (1/3)^n$, the covering will be slightly wasteful—some boxes hang over the edge of the set—but the limiting value of d is the same.

EXAMPLE 11.4.2:

A fractal that is *not* self-similar is constructed as follows. A square region is divided into nine equal squares, and then one of the small squares is selected at random and discarded. Then the process is repeated on each of the eight remaining small squares, and so on. What is the box dimension of the limiting set?

Solution: Figure 11.4.2 shows the first two stages in a typical realization of this random construction.

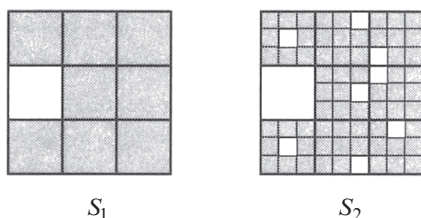


Figure 11.4.2

Pick the unit of length to equal the side of the original square. Then S_1 is covered (with no wastage) by $N = 8$ squares of side $\varepsilon = \frac{1}{3}$. Similarly, S_2 is covered by $N = 8^2$ squares of side $\varepsilon = (\frac{1}{3})^2$. In general, $N = 8^n$ when $\varepsilon = (\frac{1}{3})^n$. Hence

$$d = \lim_{\varepsilon \rightarrow 0} \frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)} = \frac{\ln(8^n)}{\ln(3^n)} = \frac{n \ln 8}{n \ln 3} = \frac{\ln 8}{\ln 3}. \blacksquare$$

Critique of Box Dimension

When computing the box dimension, it is not always easy to find a minimal cover. There's an equivalent way to compute the box dimension that avoids this problem. We cover the set with a square mesh of boxes of side ε , count the number of occupied boxes $N(\varepsilon)$, and then compute d as before.

Even with this improvement, the box dimension is rarely used in practice. Its computation requires too much storage space and computer time, compared to other

types of fractal dimension (see below). The box dimension also suffers from some mathematical drawbacks. For example, its value is not always what it should be: the set of rational numbers between 0 and 1 can be proven to have a box dimension of 1 (Falconer 1990, p. 44), even though the set has only countably many points.

Falconer (1990) discusses other fractal dimensions, the most important of which is the *Hausdorff dimension*. It is more subtle than the box dimension. The main conceptual difference is that the Hausdorff dimension uses coverings by small sets of *varying* sizes, not just boxes of fixed size ε . It has nicer mathematical properties than the box dimension, but unfortunately it is even harder to compute numerically.

11.5 Pointwise and Correlation Dimensions

Now it's time to return to dynamics. Suppose that we're studying a chaotic system that settles down to a strange attractor in phase space. Given that strange attractors typically have fractal microstructure (as we'll see in Chapter 12), how could we estimate the fractal dimension?

First we generate a set of very many points $\{\mathbf{x}_i, i = 1, \dots, n\}$ on the attractor by letting the system evolve for a long time (after taking care to discard the initial transient, as usual). To get better statistics, we could repeat this procedure for several different trajectories. In practice, however, almost all trajectories on a strange attractor have the same long-term statistics so it's sufficient to run one trajectory for an extremely long time. Now that we have many points on the attractor, we could try computing the box dimension, but that approach is impractical, as mentioned earlier.

Grassberger and Procaccia (1983) proposed a more efficient approach that has become standard. Fix a point $\mathbf{x}$ on the attractor A . Let $N_\varepsilon(\varepsilon)$ denote the number of points on A inside a ball of radius ε about $\mathbf{x}$ (Figure 11.5.1).

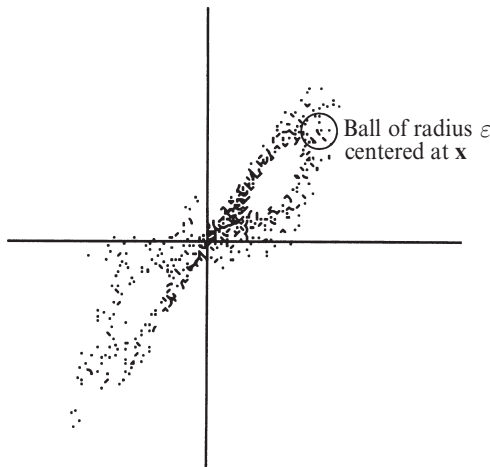


Figure 11.5.1

Most of the points in the ball are unrelated to the immediate portion of the trajectory through $\mathbf{x}$; instead they come from later parts that just happen to pass close to $\mathbf{x}$. Thus $N_{\mathbf{x}}(\varepsilon)$ measures how frequently a typical trajectory visits an ε neighborhood of $\mathbf{x}$.

Now vary ε . As ε increases, the number of points in the ball typically grows as a power law:

$$N_{\mathbf{x}}(\varepsilon) \propto \varepsilon^d,$$

where d is called the **pointwise dimension** at $\mathbf{x}$. The pointwise dimension can depend significantly on $\mathbf{x}$; it will be smaller in rarefied regions of the attractor. To get an overall dimension of A , one averages $N_{\mathbf{x}}(\varepsilon)$ over many $\mathbf{x}$. The resulting quantity $C(\varepsilon)$ is found empirically to scale as

$$C(\varepsilon) \propto \varepsilon^d,$$

where d is called the **correlation dimension**.

The correlation dimension takes account of the density of points on the attractor, and thus differs from the box dimension, which weights all occupied boxes equally, no matter how many points they contain. (Mathematically speaking, the correlation dimension involves an invariant measure supported on a fractal, not just the fractal itself.) In general, $d_{\text{correlation}} \leq d_{\text{box}}$, although they are usually very close (Grassberger and Procaccia 1983).

To estimate d , one plots $\log C(\varepsilon)$ vs. $\log \varepsilon$. If the relation $C(\varepsilon) \propto \varepsilon^d$ were valid for all ε , we'd find a straight line of slope d . In practice, the power law holds only over an intermediate range of ε (Figure 11.5.2).

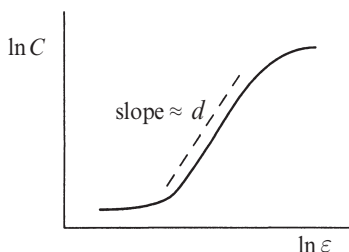


Figure 11.5.2

The curve saturates at large ε because the ε -balls engulf the whole attractor and so $N_{\mathbf{x}}(\varepsilon)$ can grow no further. On the other hand, at extremely small ε , the only point in each ε -ball is $\mathbf{x}$ itself. So the power law is expected to hold only in the *scaling region* where

(minimum separation of points on A) $\ll \varepsilon \ll$ (diameter of A).

EXAMPLE 11.5.1:

Estimate the correlation dimension of the Lorenz attractor, for the standard parameter values $r = 28$, $\sigma = 10$, $b = \frac{8}{3}$.

Solution: Figure 11.5.3 shows the results of Grassberger and Procaccia (1983). (Note that in their notation, the radius of the balls is ℓ and the correlation dimension is ν .) A line of slope $d_{\text{corr}} = 2.05 \pm 0.01$ gives an excellent fit to the data, except for large ε , where the expected saturation occurs.

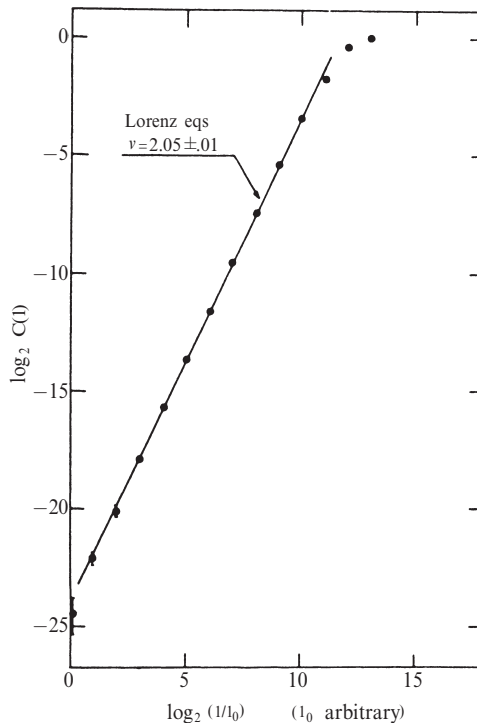


Figure 11.5.3 Grassberger and Procaccia (1983), p.196

These results were obtained by numerically integrating the system with a Runge–Kutta method. The time step was 0.25, and 15,000 points were computed. Grassberger and Procaccia also report that the convergence was rapid; the correlation dimension could be estimated to within ± 5 percent using only a few thousand points. ■

EXAMPLE 11.5.2:

Consider the logistic map $x_{n+1} = rx_n(1 - x_n)$ at the parameter value $r = r_\infty = 3.5699456 \dots$, corresponding to the onset of chaos. Show that the

attractor is a Cantor-like set, although it is not strictly self-similar. Then compute its correlation dimension numerically.

Solution: We visualize the attractor by building it up recursively. Roughly speaking, the attractor looks like a 2^n -cycle, for $n \gg 1$. Figure 11.5.4 schematically shows some typical 2^n -cycles for small values of n .

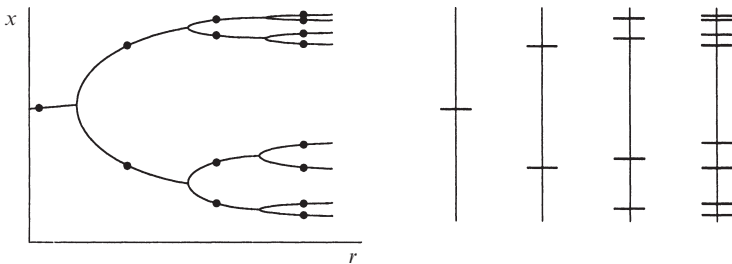


Figure 11.5.4

The dots in the left panel of Figure 11.5.4 represent the superstable 2^n -cycles. The right panel shows the corresponding values of x . As $n \rightarrow \infty$, the resulting set approaches a topological Cantor set, with points separated by gaps of various sizes. But the set is not strictly self-similar—the gaps scale by different factors depending on their location. In other words, some of the “wishbones” in the orbit diagram are wider than others at the same r . (We commented on this nonuniformity in Section 10.6, after viewing the computer-generated orbit diagrams of Figure 10.6.2.)

The correlation dimension of the limiting set has been estimated by Grassberger and Procaccia (1983). They generated a single trajectory of 30,000 points, starting from $x_0 = \frac{1}{2}$. Their plot of $\log C(\varepsilon)$ vs. $\log \varepsilon$ is well fit by a straight line of slope $d_{\text{corr}} = 0.500 \pm 0.005$ (Figure 11.5.5).

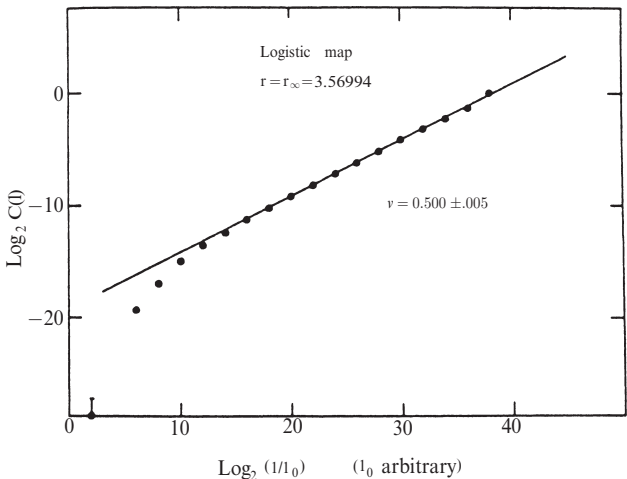


Figure 11.5.5 Grassberger and Procaccia (1983), p.193

This is smaller than the box dimension $d_{\text{box}} \approx 0.538$ (Grassberger 1981), as expected. ■

For very small ε , the data in Figure 11.5.5 deviate from a straight line. Grassberger and Procaccia (1983) attribute this deviation to residual correlations among the x_n 's on their single trajectory. These correlations would be negligible if the map were strongly chaotic, but for a system at the onset of chaos (like this one), the correlations are visible at small scales. To extend the scaling region, one could use a larger number of points or more than one trajectory.

Multifractals

We conclude by mentioning a more refined concept, although we cannot go into details. In the logistic attractor of Example 11.5.2, the scaling varies from place to place, unlike in the middle-thirds Cantor set, where there is a uniform scaling by $\frac{1}{3}$ everywhere. Thus we cannot completely characterize the logistic attractor by its dimension, or any other single number—we need some kind of distribution function that tells us how the dimension varies across the attractor. Sets of this type are called *multifractals*.

The notion of pointwise dimension allows us to quantify the local variations in scaling. Given a multifractal A , let S_α be the subset of A consisting of all points with pointwise dimension α . If α is a typical scaling factor on A , then it will be represented often, so S_α will be a relatively large set; if α is unusual, then S_α will be a small set. To be more quantitative, we note that each S_α is itself a fractal, so it makes sense to measure its “size” by its fractal dimension. Thus, let $f(\alpha)$ denote the dimension of S_α . Then $f(\alpha)$ is called the *multifractal spectrum* of A or the *spectrum of scaling indices* (Halsey et al. 1986).

Roughly speaking, you can think of the multifractal as an interwoven set of fractals of different dimensions α , where $f(\alpha)$ measures their relative weights. Since very large and very small α are unlikely, the shape of $f(\alpha)$ typically looks like Figure 11.5.6. The maximum value of $f(\alpha)$ turns out to be the box dimension (Halsey et al. 1986).

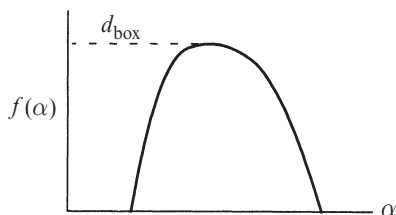


Figure 11.5.6

For systems at the onset of chaos, multifractals lead to a more powerful version of the universality theory mentioned in Section 10.6. The universal quantity is now a

function $f(\alpha)$, rather than a single number; it therefore offers much more information, and the possibility of more stringent tests. The theory's predictions have been checked for a variety of experimental systems at the onset of chaos, with striking success. See Glazier and Libchaber (1988) for a review. On the other hand, we still lack a rigorous mathematical theory of multifractals; see Falconer (1990) for a discussion of the issues.

EXERCISES FOR CHAPTER 11

11.1 Countable and Uncountable Sets

11.1.1 Why doesn't the diagonal argument used in Example 11.1.4 show that the rationals are also uncountable? (After all, rationals can be represented as decimals.)

11.1.2 Show that the set of odd integers is countable.

11.1.3 Are the irrational numbers countable or uncountable? Prove your answer.

11.1.4 Consider the set of all real numbers whose decimal expansion contains only 2's and 7's. Using Cantor's diagonal argument, show that this set is uncountable.

11.1.5 Consider the set of integer lattice points in three-dimensional space, i.e., points of the form (p, q, r) , where p , q , and r are integers. Show that this set is countable.

11.1.6 ($10x \bmod 1$) Consider the decimal shift map $x_{n+1} = 10x_n \bmod 1$.

- Show that the map has countably many periodic orbits, all of which are unstable.
- Show that the map has uncountably many aperiodic orbits.
- An "eventually-fixed point" of a map is a point that iterates to a fixed point after a finite number of steps. Thus $x_{n+1} = x_n$ for all $n > N$, where N is some positive integer. Is the number of eventually-fixed points for the decimal shift map countable or uncountable?

11.1.7 Show that the binary shift map $x_{n+1} = 2x_n \bmod 1$ has countably many periodic orbits and uncountably many aperiodic orbits.

11.2 Cantor Set

11.2.1 (Cantor set has measure zero) Here's another way to show that the Cantor set has zero total length. In the first stage of construction of the Cantor set, we removed an interval of length $\frac{1}{3}$ from the unit interval $[0, 1]$. At the next stage we removed two intervals, each of length $\frac{1}{9}$. By summing an appropriate infinite